FR3 UAV Channel Knowledge Maps: A Dataset and an Interpretable Attenuation Model

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***Abstract*—This paper introduces a dense FR3 channel knowledge map dataset for unmanned aerial vehicle base stations. It contains 16180 ray-traced realizations from 3236 locations across 66 cities at 7.125 GHz, with attenuation, geometric visibility, delay spread, and angular spread stored at 513×513 resolution for five transmitter heights per location. We use attenuation as a focused first benchmark and propose a height-aware, auditable model. Its LoS branch combines free-space loss with a coherent image-source two-ray term, whereas its NLoS branch combines a COST231-shaped basis with local morphology and topology-specific linear calibration. The dataset-supplied ray-traced visibility mask selects the branch. Under a strict city holdout, the frozen model reaches 1.9277 dB overall root mean square error, with 1.7370 dB in LoS and 3.5275 dB in NLoS, on 2590 maps from 14 unseen cities. Its median CPU time is 0.0602 s per map. The result provides a transparent attenuation baseline and a physical anchor for later distribution-aware refinement of all three channel quantities.**

***Index Terms*—channel attenuation, channel knowledge map, delay spread, path loss, radio map, unmanned aerial vehicle.**

I. Introduction

Uncrewed aerial vehicles (UAVs) carrying base stations can support future 6G networks through mobility and an increased probability of line of sight (LoS). Their deployment nevertheless requires fast location-dependent channel information for placement and resource management. Channel knowledge maps (CKMs) address this need by indexing channel information by transmitter or receiver location [1]. A CKM may store attenuation, multipath delays and angles, or richer channel descriptions.

CKMs can be obtained from measurements or channel models. Measurements are accurate but costly to collect densely, whereas deterministic ray tracing (RT) offers spatially complete labels at substantial computational cost. Available CKM datasets commonly rely on RT [2], [3], and learned surrogates such as RadioUNet and PMNet use convolutional encoder–decoders to predict dense channel maps from city geometry and transmitter information [4], [5].

The upper mid-band frequency range FR3, between 7 and 24 GHz, is a candidate for combining wider bandwidth than FR1 with more favorable coverage than FR2 [6]. It is especially relevant to elevated transmitters, which often benefit from LoS yet must still operate through urban NLoS regions. Assessing that setting requires channel data across varied building morphologies and transmitter heights. To our knowledge, no public CKM dataset currently provides this combination in FR3.

a) Limitations of the existing CKM datasets: Table I compares dataset scale, transmitter elevation, released channel quantities, geometric inputs, and ready-to-use topology labels. "Building-height raster included" means that the released learning input preserves a spatially varying height for each building rather than only a binary footprint; it does not imply that every source height was directly surveyed. Such height information is sparse in OpenStreetMap [8]. We therefore use global building-scale products that provide both footprint and height [9], [10].

b) Limitations of the existing CKM modeling approaches: Recent CKM estimators use increasingly expressive convolutional, transformer, variational, and diffusion architectures. Such models are valuable when sparse observations or unresolved propagation effects must be inferred, but their added complexity is rarely isolated from stable propagation structure. This comparison is particularly relevant when geometric visibility is already available and transmitter height varies continuously. A calibrated analytical base can explain the repeatable structure before a learned model is asked to represent local residuals and multimodal tails.

c) Contributions: The paper makes two contributions:

1) We introduce an FR3 CKM dataset generated by RT at 3236 locations across 66 cities. Five transmitter heights per location yield 16180 dense maps containing attenuation, LoS, delay spread, and angular spread. The labeled release also includes topology classes for attenuation and spread modeling, plus a transmitter-height bin for every realization. Global building footprint and height products provide the underlying 3D geometry [9], [10].
2) We establish an attenuation benchmark through a height-aware model that separates LoS and NLoS mechanisms. Coherent two-ray structure handles LoS, while topology-specific linear calibration corrects a COST231-shaped NLoS basis. Every fitted quantity is frozen before evaluation on unseen cities.

Section II defines the data and evaluation contract, Section III presents the attenuation model, Section IV reports the results, and Section V concludes the paper.

II. Data and Evaluation Contract

Each sample is a 513×513 urban CKM with 1 m spatial resolution. One elevated transmitter is horizontally centered in the map, operates at 7.125 GHz, and has a sample-dependent height $h_{\mathrm{tx}} \approx 12\,\mathrm{m}$ to $478\,\mathrm{m}$. Every non-building pixel is a

TABLE I
COMPARISON OF DENSE CKM DATASETS.

Property	This work	UrbanRadio3D [7]	RadioMapSeer [2]	CKMImageNet [3]
Variable elevated Tx height	yes	no	no	no
Distinct spatial layouts	3236	701	701	42
Map size	513×513	256×256	256×256	128×128
Horizontal resolution	1 m	1 m	1 m	2 m
Channel quantities	Att. (PL), LoS, DS, AS	PL, DoA, ToA	PL, ToA	Gain, AoA/AoD, delay
Building-height raster included	yes	yes	no	no
Per-sample topology labels	3 and 6 classes	no	no	no

receiver candidate at $h_{\rm rx} = 1.5\,$m. The HDF5 stores dense attenuation, delay spread in ns, angular spread in degrees, and geometric LoS for every realization. All three channel quantities are released; the model and benchmark in this paper deliberately focus on attenuation, the primary link-budget quantity and the one for which the dominant propagation structure can be isolated most directly.

The released artifact CKM_Dataset_270326_labeled.h5 stores `topology_3_class`, `topology_6_class`, and `antenna_bin` per realization. The three- and six-class fields support attenuation and spread routing, respectively, while the antenna field records the low, middle, or high transmitter-height bin. All labels are deterministic from building density, mean nonzero building height, and transmitter height; they use neither city identity nor channel targets.

Fig. 1 illustrates the spatial alignment and scalar height metadata of one released realization.

The attenuation metric first excludes all building pixels and then retains only finite ray-traced targets in the stored $[20, 185]$ dB support. The target is an unsigned 8-bit array with 1 dB quantization.

Headline metrics use the binary `LineOfSight` field emitted by the configured MATLAB ray tracer and stored as `los_mask`. It is keyed by the city geometry, transmitter position and height, receiver grid, and ray-tracing settings, and is not inferred by the attenuation prior. Building pixels are removed before this mask defines LoS and NLoS receiver sets.

The split is made by city rather than by pixel, tile, or random sample. The official split contains 10840 maps from 37 training cities, 2750 maps from 15 validation cities, and 2590 maps from 14 final test cities. No city is moved between these partitions. All lookup tables, standardization statistics, and regression coefficients use training cities only. Validation is used for development checks; headline results and uncertainty intervals use the final test partition after every fitted quantity has been frozen.

For valid-receiver mask $m_i(p)$, prediction $\hat{y}_i(p)$, and target $y_i(p)$, the pixel-weighted test error is

$$\mathrm{RMSE} = \sqrt{\frac{\sum_i \sum_p m_i(p)[\hat{y}_i(p) - y_i(p)]^2}{\sum_i \sum_p m_i(p)}}. \tag{1}$$

We also report mean absolute error (MAE) and an unweighted macro average of per-city RMSE. RMSE remains the primary pixel-weighted, unit-aware metric. Uncertainty is estimated by 20000 percentile-bootstrap draws over the 14 final test cities.

TABLE II
EVALUATION SETTING.

Item	Setting
Map	513×513, 1 m pixels
Carrier	7.125 GHz
Transmitter	Centered; variable height
Receiver	Ground, 1.5 m; buildings masked
Visibility	Configured-ray-tracer LoS output stored in HDF5
Split	10840/2750/2590 train/validation/test maps by city
Test cities	Barcelona, Beijing, Carcassonne, Copenhagen, Halifax, Jaipur, Johannesburg, Key West, Kuala Lumpur, Osaka, Phuket, Pune, Surat, Vancouver

Each draw samples cities with replacement and recomputes the pixel-weighted metric; this preserves within-city spatial dependence rather than treating hundreds of millions of pixels as independent observations.

III. Height-Aware Attenuation Prior

Fig. 2 summarizes calibration and deployment. The method uses known geometry to compute distances, a building/visibility mask to route LoS and NLoS pixels, and local windows over the building-height map. Only compact calibration tables and linear weights are learned. They are frozen before a held-out city is evaluated.

A. LoS Branch

Let d_{2D} be the horizontal transmitter-receiver distance. The direct and image-source reflected distances are

$$\begin{aligned} d_{\rm LoS} &= \sqrt{d_{2D}^2 + (h_{\rm tx} - h_{\rm rx})^2}, \\ d_{\rm ref} &= \sqrt{d_{2D}^2 + (h_{\rm tx} + h_{\rm rx})^2}. \end{aligned} \tag{2}$$

The free-space term follows the Friis distance dependence, while the coherent correction retains the phase difference between the direct and reflected paths [11]:

$$\mathrm{FSPL} = 32.45 + 20\log_{10}(d_{\rm LoS}/1000) + 20\log_{10}(f_{\rm MHz}), \tag{3}$$

$$C_{\rm 2ray} = -20\log_{10}\left|1 + \rho(h_{\rm tx})\frac{d_{\rm LoS}}{d_{\rm ref}} e^{-j[k(d_{\rm ref}-d_{\rm LoS})+\phi(h_{\rm tx})]}\right|. \tag{4}$$

Here $k = 2\pi f/c$ is the free-space wavenumber, ρ is an effective reflected-field amplitude, and ϕ is a fitted phase offset. The latter two are CKM calibration terms, not material Fresnel coefficients. The deployed LoS prior is

$$\mathrm{PL}_{\rm LoS} = \mathrm{FSPL} + C_{\rm 2ray} + \beta_{\rm LoS}(h_{\rm tx}) + r(d_{2D}, h_{\rm tx}). \tag{5}$$

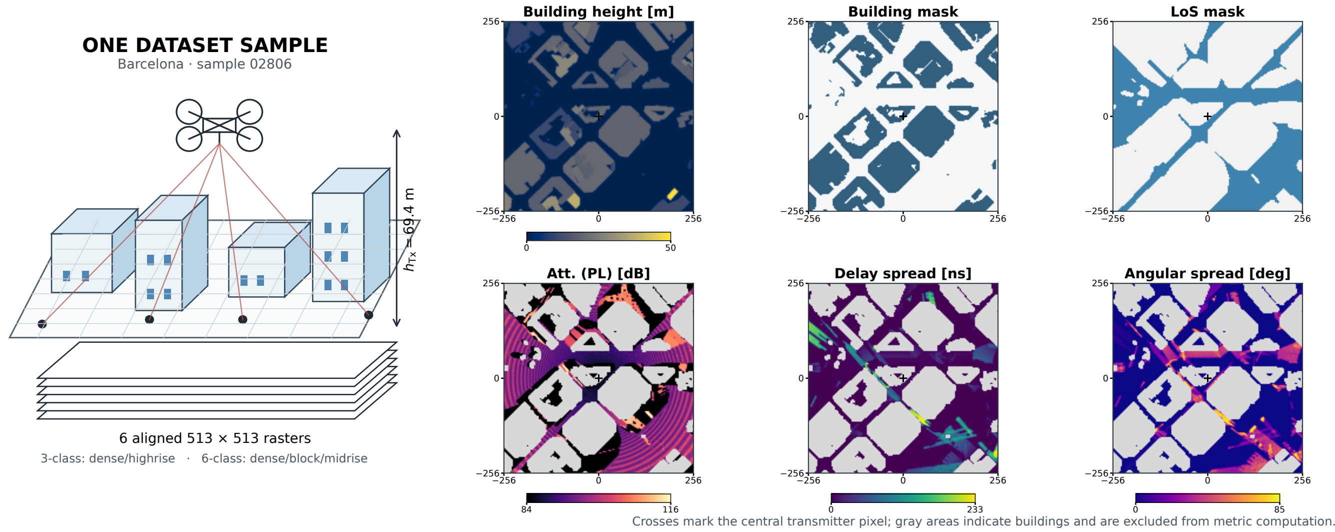

Fig. 1. One released Barcelona realization at $h_{\rm tx} = 69.4\,\text{m}$. The left schematic separates transmitter height from the six co-registered rasters; the panels show the stored building height, building mask, LoS, attenuation, delay spread, and angular spread. Crosses mark the centered transmitter. Gray areas indicate buildings and are excluded from metric computation.

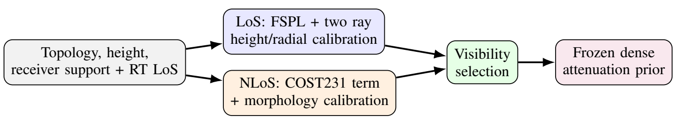

Fig. 2. The prior keeps LoS and NLoS propagation mechanisms separate. All CKM-specific calibration is fitted on training cities and then frozen.

Calibration uses valid LoS pixels from calibration-side cities only. In each 5 m height bin, a small grid search selects effective amplitude and phase. A closed-form bias removes the remaining vertical offset. Finally, a smoothed radial table stores the mean residual left by the coherent model. Neighboring height bins are smoothed and linearly interpolated at inference. This structured sequence targets the constructive/destructive rings visible around the centered transmitter without giving the prior enough flexibility to memorize building layouts.

In implementation terms, the LoS correction is a lookup table. The stored entries are the effective ρ, ϕ, and bias curves indexed by transmitter-height bin, together with r indexed by height and radial-distance bin. Inference linearly interpolates neighboring height entries and reads the radial correction for each receiver; it does not refit on the new map. This distinction matters: although the table is data-calibrated, all 14 reported test cities are unseen during table construction. The free-space/two-ray structure supplies extrapolatable geometry, while the lookup table supplies a compact correction learned on other environments.

For each height bin, candidates $\hat{\rho} \in \{0, 0.05, \ldots, 1.5\}$ and 48 uniformly spaced phases in $[-\pi, \pi)$ are evaluated. For a candidate pair, the remaining bias is

$$\hat{\beta}(\hat{\rho}, \hat{\phi}) = \frac{1}{|B_h|} \sum_{i \in B_h} \left[y_i - \mathrm{FSPL}_i - C_{2\mathrm{ray},i}(\hat{\rho}, \hat{\phi}) \right]. \tag{6}$$

The pair minimizing the bin RMSE is retained. After smoothing the height curves, the remaining error is averaged by radius and stored as r. This ordering prevents the table from encoding arbitrary two-dimensional building texture.

B. NLoS Branch

The blocked-link branch starts from a COST231/Hata-derived urban trend [12]. Its source domain does not match this 7.125 GHz CKM, so the expression is used as an inspectable basis function rather than a transferred final law. Each map has 513×513 cells at $1\,\text{m}$ spacing, with the transmitter projection at cell $(256, 256)$. For a receiver cell $p = (u, v)$, the implementation first computes

$$\begin{aligned} d_{2D}(p) &= 1\,\text{m}\sqrt{(u-256)^2 + (v-256)^2}, \\ d(p) &= \max\{d_{2D}(p), 1\,\text{m}\}, \\ h_t &= \max\{h_{\rm tx}, 1\,\text{m}\}, \quad h_r = \max\{h_{\rm rx}, 0.5\,\text{m}\} = 1.5\,\text{m}. \end{aligned} \tag{7}$$

Thus $d_{\rm km} = \max\{d(p)/1000\,\text{m}, 0.001\}$; all logarithms below receive the numerical value in the stated unit. With $a(h_{\rm rx}) = (1.1\log_{10} f_{\rm MHz} - 0.7)h_{\rm rx} - (1.56\log_{10} f_{\rm MHz} - 0.8)$, and $\eta(h_t) = 44.9 - 6.55\log_{10} h_t$, define

$$\begin{aligned} \widetilde{\mathrm{PL}}_C = {} & 46.3 + 33.9\log_{10} f_{\rm MHz} - 13.82\log_{10} h_t - a(h_r) \\ & + \eta(h_t)\log_{10} d_{\rm km} + 3. \end{aligned} \tag{8}$$

The $+3$ dB metropolitan correction is retained inside this empirical basis; Eq. (8) is not claimed to be a directly valid COST231 deployment law at the carrier frequency and heights considered here. The regression input is the clipped map

$$\mathrm{PL}_C = \mathrm{clip}\left(\widetilde{\mathrm{PL}}_C, 0, 185\right). \tag{9}$$

The clipping is applied before PL_C enters the regression. For the elevation-dependent morphology features below, let $\theta(p) = (180/\pi)\tan^{-1}[(h_t - h_r)/d(p)]$.

The visibility mask identifies that a direct path is blocked but not which street canyon, roof edge, or clutter configuration dominates the remaining path. We therefore augment PL_C with exactly defined local quantities. Let $B(p) = \mathbf{1}\{T(p) \neq 0\}$ be the building indicator from topology height T; let $G = 1 - B$; and let $N(p) = \mathbf{1}\{m_{\mathrm{LoS}}(p) \leq 0.5\}G(p)$ be the NLoS-ground indicator. For odd k, define the reflect-padded box mean

$$\mathcal{A}_k[z](p) = \frac{1}{k^2} \sum_{q \in \mathcal{W}_k(p)} z(q). \tag{10}$$

The implemented features are

$$\begin{aligned} &\ell_d = \ln(1 + d_{2D}/1\,\mathrm{m}), && \delta_k = \mathcal{A}_k[B], \\ &h_k = \mathcal{A}_k[TB]/90\,\mathrm{m}, && n_k = \mathcal{A}_k[N], \\ &\sigma_{\mathrm{sh}} = 2.3197[\max(90 - \theta, 0)]^{0.2361}, && \\ &\theta_n = \mathrm{clip}(\theta/90, 0, 1), && k \in \{15, 41\}. \end{aligned} \tag{11}$$

The shadowing proxy is an empirical elevation feature; its role here is deterministic feature construction, not stochastic shadow-fading generation. Thus, h_k is the window-averaged building height divided by 90 m, including ground pixels as zeros; it is not the mean conditioned only on buildings. The final vector, in implementation order, is

$$\begin{aligned} \mathbf{x} = [&\mathrm{PL}_C, \ell_d, \delta_{15}, \delta_{41}, h_{15}, h_{41}, \delta_{41}\ell_d, \\ &n_{15}, n_{41}, n_{41}\ell_d, \sigma_{\mathrm{sh}}, \theta_n, n_{41}\theta_n, 1]^\top. \end{aligned} \tag{12}$$

Table III removes any remaining ambiguity by giving the array operation, scale, and interpretation of every regressor. Window sizes are in pixels and therefore metres here; $\mathcal{A}_k$ always includes all k^2 cells after reflect padding, including zeros over ground in the height features. Training standardizes non-intercept features within each fitted regime; coefficients are then transformed back to the raw feature space for the deployed dot product.

The regression regime is observable. From a complete topology map, let D be its building-pixel fraction and H its mean nonzero building height. The class is dense/high-rise if $D \geq 0.2549$ or $H \geq 15.95$ m; open/low-rise if $D \leq 0.1957$ and $H \leq 10.91$ m; and mixed/mid-rise otherwise. Transmitter-height bins are low for $h_{\mathrm{tx}} \leq 58.12$ m, middle up to 103.85 m, and high above it. For each training map, at most 1024 NLoS pixels are selected with a deterministic sample seed (global seed 1234). Let $\mathbf{z}$ contain the standardized non-intercept features and an unpenalized intercept. For regime q, the fitted ridge model is

$$\begin{aligned} &\hat{\mathbf{w}}_q = \arg\min_{\mathbf{w}} \sum_{i \in q} (\mathbf{z}_i^\top \mathbf{w} - y_i)^2 + 10^{-2}\|\mathbf{w}_{\neg b}\|_2^2, \\ &\mathrm{PL}_{\mathrm{NLoS}} = \mathrm{clip}(\mathbf{x}^\top \hat{\mathbf{w}}_q, 20, 185). \end{aligned} \tag{13}$$

In the second line, $\hat{\mathbf{w}}_q$ denotes the algebraically equivalent raw-space coefficients after undoing standardization. All nine topology and height regimes are fitted on the 10840 training maps only. Table IV gives representative coefficients for the middle transmitter-height bin. Their magnitudes must be read together with the feature scales in Table III; they are not feature importance scores. The accompanying CSV contains all nine 14-term vectors at machine precision.

The topology split changes the fitted relationship, not only its intercept. At middle height, the COST231 coefficient changes sign between open and the other two classes, while the distance, density, local-visibility, and elevation coefficients also change substantially. Correlated raw coefficients cannot be ranked as independent feature importances, but these cross-topology changes show that one global set of slopes would combine distinct propagation regimes.

The current artifact contains all 3×3 NLoS topology/height vectors; the inference code checks compatible height-bin fallbacks only if an exact key is absent. This branch is semi-empirical, but its raw estimate, each feature, selected regime, coefficient, and pre-clipping contribution remain auditable.

The complete prior uses the deterministic LoS mask m_{LoS}:

$$\mathrm{PL}_{\mathrm{prior}} = m_{\mathrm{LoS}}\mathrm{PL}_{\mathrm{LoS}} + (1 - m_{\mathrm{LoS}})\mathrm{PL}_{\mathrm{NLoS}}. \tag{14}$$

Both deployed branches are finally clipped to the target support $[20, 185]$ dB.

IV. Results

The component ablation uses the 2750-map validation partition; every fitted variant still uses training cities only. The coherent term reduces LoS RMSE from 3.7900 to 1.7412 dB; the radial table gives a smaller further reduction. The raw COST231 term is not accurate enough by itself. A global NLoS ridge model produces a strong reduction, and observable topology and height regimes improve it further. On final test, topology-only routing retains over 90% of the NLoS RMSE reduction obtained by moving from one global calibration to all nine topology/height regimes; topology is therefore the dominant partition variable, with transmitter-height bins providing a smaller refinement. A topology-specific intercept is already a usable coarse predictor; the heterogeneous slopes supply the remaining gain. A separately recalibrated eight-term variant retaining the 41-pixel context, geometry, visibility, COST231, and bias terms also recovers over 90% of the complete model's improvement over a constant regime baseline.

Table V gives the final attenuation-only test result. The prior reaches 1.9277 dB RMSE and 1.2151 dB MAE across all valid receiver pixels. The LoS error is 1.7370 dB. NLoS remains the hard regime at 3.5275 dB, consistent with the unobserved identity of the dominant diffracted or reflected path after direct-path blockage.

Per-city RMSE ranges from 1.6097 dB in Halifax to 2.5827 dB in Osaka; the unweighted 14-city mean is 1.9823 dB. This spread and the city-bootstrap interval prevent the global score from being read as uniform performance across environments.

After excluding buildings and ground pixels without a valid target, the test metric covers 423 087 119 receivers, of which 7.413% are NLoS. Although the pixel-weighted score is therefore LoS dominated, 424 of the 2590 test maps contain more than 25% NLoS receivers, motivating the separate regime metrics.

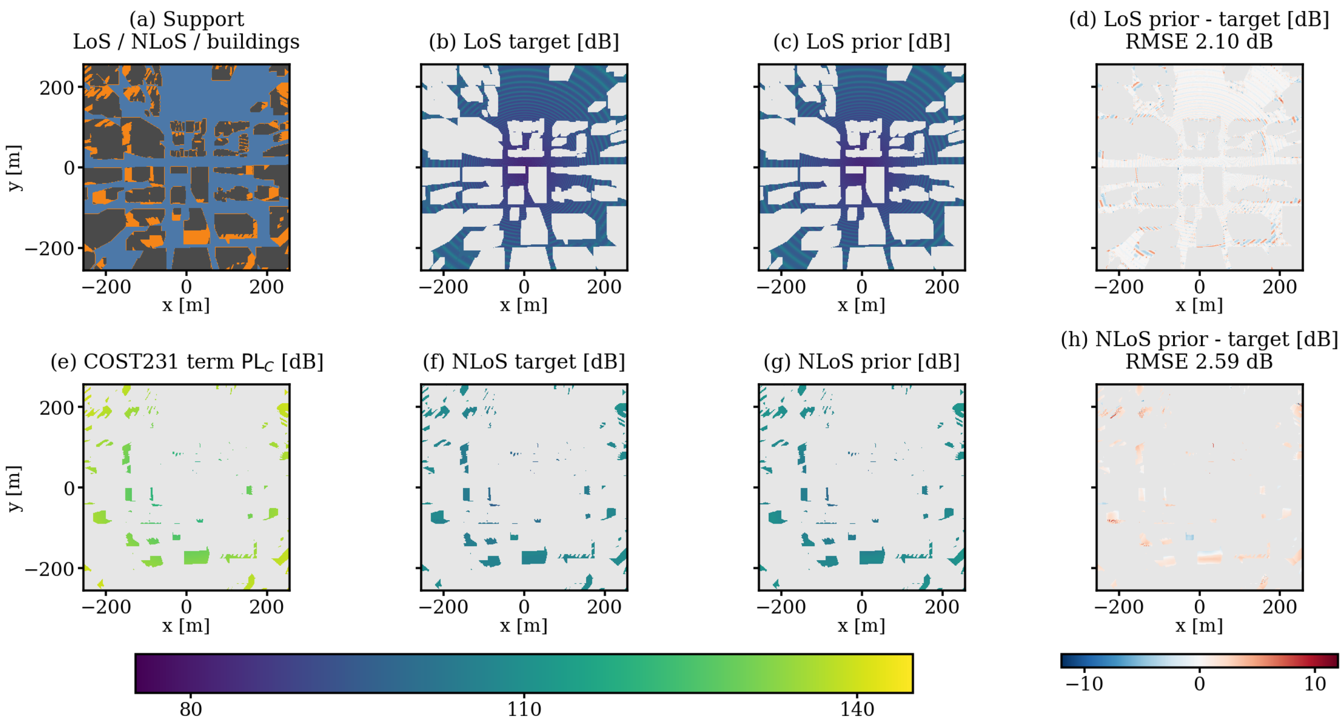

Fig. 3. A held-out example from Vancouver ("sample_15262", $h_{\text{tx}} = 50.30\,\text{m}$, dense/high-rise, low-transmitter regime). Panel (a) shows the geometric support; (b)–(d) are the LoS target, frozen lookup-table prior, and signed error; (e) visualizes the raw COST231 term PL_C; and (f)–(h) give the NLoS target, calibrated prior, and signed error. Grey excludes pixels outside the displayed regime or without a valid target. The unseen map contains 119180 LoS and 25607 NLoS receivers, with 2.10 dB and 2.59 dB RMSE, respectively.

TABLE III
EXACT CONSTRUCTION OF EVERY NLOS MULTILINEAR-REGRESSION INPUT AT RECEIVER CELL p. THE FEATURE ORDER IS ALSO THE COEFFICIENT-VECTOR ORDER. "UNITLESS" MEANS THAT THE STATED METRE OR DEGREE NORMALIZATION HAS ALREADY BEEN APPLIED.

j	Feature $x_j(p)$	Exact computation	Scale and role
1	PL_C	Eq. (9)	dB; clipped COST231 term
2	ℓ_d	$\ln[1 + d_{2D}(p)/1\,\text{m}]$	unitless; radial growth
3	δ_{15}	$15^{-2}\sum_{q\in\mathcal{W}_{15}(p)} \mathbf{1}\{T(q) \neq 0\}$	unitless; local building fraction
4	δ_{41}	$41^{-2}\sum_{q\in\mathcal{W}_{41}(p)} \mathbf{1}\{T(q) \neq 0\}$	unitless; contextual building fraction
5	h_{15}	$(15^2\,90\,\text{m})^{-1}\sum_{q\in\mathcal{W}_{15}(p)} T(q)\mathbf{1}\{T(q) \neq 0\}$	unitless; local height mass
6	h_{41}	$(41^2\,90\,\text{m})^{-1}\sum_{q\in\mathcal{W}_{41}(p)} T(q)\mathbf{1}\{T(q) \neq 0\}$	unitless; contextual height mass
7	$\delta_{41}\ell_d$	$\delta_{41}(p)\ell_d(p)$	unitless; density–range interaction
8	n_{15}	$15^{-2}\sum_{q\in\mathcal{W}_{15}(p)} \mathbf{1}\{m_{\text{LoS}}(q) \leq 0.5\}\mathbf{1}\{T(q) = 0\}$	unitless; local blocked-ground fraction
9	n_{41}	$41^{-2}\sum_{q\in\mathcal{W}_{41}(p)} \mathbf{1}\{m_{\text{LoS}}(q) \leq 0.5\}\mathbf{1}\{T(q) = 0\}$	unitless; contextual blocked-ground fraction
10	$n_{41}\ell_d$	$n_{41}(p)\ell_d(p)$	unitless; blockage–range interaction
11	σ_{sh}	$2.3197[\max\{90 - \theta(p), 0\}]^{0.2361}$	dB; elevation-dependent shadowing proxy
12	θ_n	$\text{clip}[\theta(p)/90, 0, 1]$	unitless; normalized elevation angle
13	$n_{41}\theta_n$	$n_{41}(p)\theta_n(p)$	unitless; blockage–elevation interaction
14	1	an all-ones array with the same 513×513 shape	unitless; intercept

Table VI complements the raw coefficients with their actual test-set scale. For each NLoS receiver, its regime-specific coefficient is multiplied by its feature value before clipping; the table reports the signed mean of that term over all valid NLoS receivers. Opposite signs across regimes can cancel in this average, so the values are scale diagnostics rather than causal feature importance scores.

Height is part of the estimator rather than a post-hoc reporting variable. Table VII shows the frozen prior on the final test cities only for four physical height intervals. Error decreases from 2.1876 dB at 12 m to 50 m to 1.5591 dB above 300 m. The largest change is in NLoS, where low transmitters interact more strongly with local rooftops and street canyons. The trend does not imply that altitude alone solves dense prediction; it shows why a single height-independent calibration would combine physically different regimes.

The result is best interpreted as a strong physical/statistical anchor rather than a universal propagation model. The coefficients are tied to the fixed frequency, antenna assumptions, simulator, and available visibility mask. The city split tests

TABLE IV
SELECTED RAW-SPACE NLOS COEFFICIENTS FOR THE MIDDLE TRANSMITTER-HEIGHT BIN.

Feature	Open/low-rise	Mixed/mid-rise	Dense/high-rise
PL_C	-0.117224	0.109671	0.117467
ℓ_d	8.669730	4.265489	3.320933
δ_{41}	-14.501030	-29.259829	-33.524686
n_{41}	15.473961	11.327532	7.311767
σ_{sh}	-10.152455	-7.352513	-11.289259
θ_n	-20.718816	-13.965305	-22.991302
Bias	144.446889	119.161245	150.024085

TABLE V
FROZEN PRIOR ON THE 2590-MAP, 14-CITY TEST SET. INTERVALS RESAMPLE CITIES, NOT PIXELS.

Metric	Estimate	95% interval
Overall RMSE [dB]	1.9277	[1.7960, 2.0967]
Overall MAE [dB]	1.2151	[1.1350, 1.3174]
LoS RMSE [dB]	1.7370	[1.6603, 1.8225]
NLoS RMSE [dB]	3.5275	[3.2106, 3.8529]
Macro city RMSE [dB]	1.9823	[1.8484, 2.1293]

TABLE VI
PIXEL-WEIGHTED SIGNED MEAN NLOS REGRESSION CONTRIBUTIONS OVER ALL 31 365 314 VALID NLOS TEST RECEIVERS. ROWS ARE ORDERED BY ABSOLUTE VALUE.

Feature	Mean contribution [dB]
Bias	149.774
σ_{sh}	-72.725
ℓ_d	20.188
PL_C	13.826
δ_{41}	-6.287
$\delta_{41}\ell_d$	4.948
θ_n	-4.454
n_{41}	1.771

TABLE VII
PRIOR RMSE BY HEIGHT ON THE 14-CITY FINAL TEST SET.

Height	Overall	LoS	NLoS	Maps
12–50 m	2.1876	1.7943	3.8133	624
50–150 m	1.9538	1.7788	3.4256	1385
150–300 m	1.6557	1.6370	2.3584	387
300–500 m	1.5591	1.5496	2.2827	194

transfer across urban layouts, but the labels still come from ray tracing rather than field measurements.

A. Limitations, Visibility, and Computational Comparison

The visibility mask is the configured MATLAB ray tracer's own binary `LineOfSight` output. Identical city geometry, transmitter position and height, receiver grid, and propagation settings therefore define one mask that can be stored and reused. In a repeated five-condition CPU audit, all 742 392 receiver visibility values were identical between runs.

"State" denotes stored entries plus coefficients for the prior and parameters for the networks. Its final 14-term NLoS dot product averages exactly $(31\,365\,314/2590)14 = 0.170 \times 10^6$ MAC, with a 3.68×10^6-MAC full-grid upper bound; local statistics, nonlinear functions, lookups, and routing are additional operations.

Neural medians use official code, batch one, six CPU threads, 10 warmups, and 50 timed forward passes; MACs cover all executed convolutions. Input and I/O are excluded. The 100-map prior timing includes feature construction and LoS evaluation; separate HDF5 reading has a 0.0046 s median. Despite their native 256^2 grids, RadioUNet and PMNet take $2.2\times$ and $6.7\times$ the prior time.

MATLAB R2024b takes a median 102.289 s (range 91.452 s to 260.310 s) over five Barcelona layouts at 1 m resolution, with one reflection and no diffraction. It also generates visibility and all three quantities, so the raw $1700\times$ gap is not a like-for-like speedup. A future journal paper will enable a closer multi-output comparison through spread priors and distribution-aware refinement.

V. CONCLUSION

This paper introduced a dense FR3 UAV CKM dataset with attenuation, delay and angular spread, and geometric visibility for 16180 realizations from 66 cities. Its labeled HDF5 release includes three- and six-class topology partitions and transmitter-height bins. The attenuation benchmark assigns stable structure to coherent radial and two-ray LoS behaviour, COST231 and local NLoS morphology, and deterministic visibility routing. All calibration is frozen before unseen cities are evaluated.

On 2590 maps from 14 held-out cities, the prior reaches 1.9277 dB overall RMSE, 1.7370 dB LoS RMSE, and 3.5275 dB NLoS RMSE. The component study identifies coherent LoS structure and topology-partitioned NLoS calibration as the central ingredients. Topology is the dominant NLoS partition variable: topology-only routing nearly matches the full topology/height calibration, and the fitted slopes change markedly across urban classes. Separate regime reporting remains essential because the aggregate metric is LoS dominated while hundreds of maps contain substantial NLoS support.

The broader conclusion is that CKM estimators should explain stable structure before learning local detail. This yields an auditable 0.0602 s CPU surrogate and a frozen anchor for residual learning. A future journal paper will extend the analytical bases to delay and angular spread, then use deep learning to refine all three quantities and model their remaining output distributions. Non-flat terrain and transfer across frequencies and antenna configurations are further priorities.

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TABLE VIII

CPU BENCHMARKS AND OPERATION COUNTS. ALL TIMINGS USE THE SAME AMD RYZEN 5 5600X WITH GPU EXECUTION DISABLED.

Method	Grid	State	Conv. MAC	CPU [s]
Proposed prior	513^2	66.8k+126	0.170M*	0.0602
RadioUNet [4]	256^2	13.27M	12.73G	0.1339
PMNet [5]	256^2	33.34M	52.71G	0.4042
MATLAB RT	513^2	–	–	102.289

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